

UNIVERSITÄT
LEIPZIG

From visual realism to functional model understanding

Polarization representations and the Poincaré sphere

Jonas Bley

Study basis: preregistration, osf.io/u6xk9

Overview

1. Mental models and model understanding
2. Scope of model understanding
3. The polarization applet
4. Hypotheses and study design
5. Analysis and current status

Mental models and model understanding

Gestalt and Functionality

Gestalt

Forms, spatial arrangements, and structures in a learner's mental representation.

Image placeholder: Gestalt
Your chosen visual example

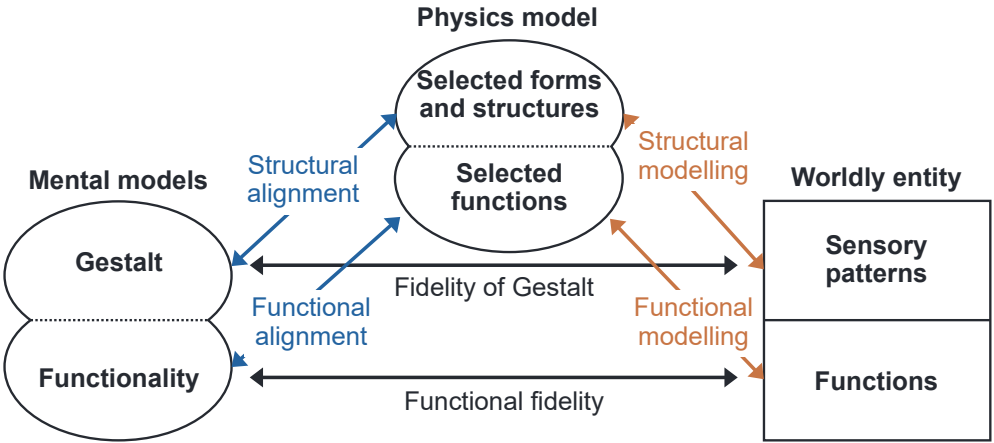
Functionality

Relationships and operations used to explain behavior and make predictions.

Image placeholder: Functionality
Your chosen functional example

Mental-model dimensions: (Ubben and Bitzenbauer, 2022); supplied terminology report.

Mental models, scientific models, and the world



Adapted from the supplied theoretical diagram. Black: fidelity; blue: expert alignment; orange: scientific modelling.

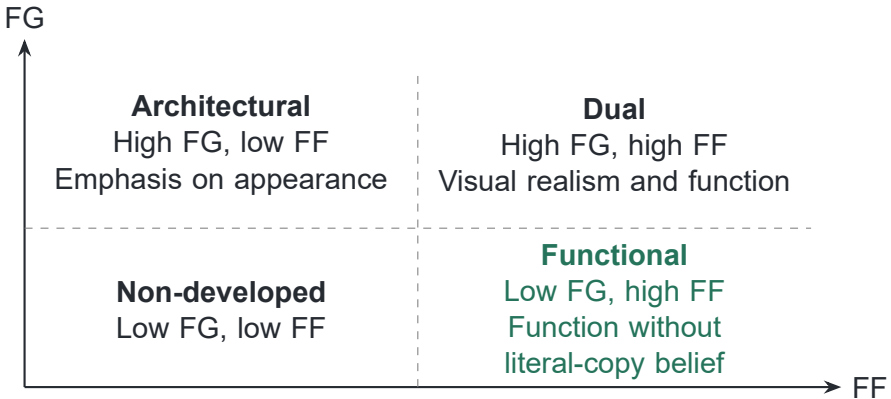
Fidelity beliefs within model understanding

	Fidelity of Gestalt (FG)	Functional Fidelity (FF)
Learner's belief	How closely does my model depict what the phenomenon looks like?	How adequately does my model describe how the phenomenon works?
High expression	Strong belief in sensory or visual faithfulness.	Strong belief in functional adequacy.
Interpretation	May encourage literal readings of abstract images.	Can support valuing prediction and explanation.

FF and FG capture fidelity beliefs. FA and GA capture expert alignment within model understanding.

(Ubben and Bitzenbauer, 2022); examples and application to polarization adapted for this study.

Four profiles of model understanding



Conceptual profiles after (Ubben and Bitzenbauer, 2022). The study measures continuous scores; these are not fixed learner categories.

Fidelity and alignment answer different questions

Fidelity: the learner's stance

- What does the learner believe the model is faithful to?
- FG: appearance; FF: function.

Alignment: expert correspondence

- **Structural alignment (StA):** correspondence with expert-selected forms and structures.
- **Functional alignment (FA):** correspondence with expert-selected functions.

A learner can reproduce the correct sphere diagram while believing it is a physical sphere surrounding the light beam.

High structural alignment can coexist with a literal interpretation.

Terminology from the supplied synthesis. Structural alignment is denoted GA (Gestalt alignment) in this study.

Scope of model understanding

Model understanding and model competencies

Model understanding in this study

Fidelity beliefs (FG, FF): perceived faithfulness to appearance and function.

Expert alignment (GA, FA): correspondence with expert-selected structures and functions.

Model competencies concern the reflective use, construction, comparison, testing, and revision of models.

Scope of the measurements

The study measures model understanding, without directly assessing model competencies.

Supplied terminology synthesis; modeling practices and metaknowledge: (Schwarz et al., 2009).

Two pathways for an additional visualization

Representational support

- Make state relationships visible.
- Connect notation, geometry, and optical transformations.
- Support expert-aligned explanations and predictions.

Spontaneous reflection

- Encounter different representations of the same state.
- Reconsider literal interpretations.
- Attend to shared functions and limits.

Expert alignment may improve even when fidelity beliefs remain stable. Both contribute to model understanding.

Multiple representations: (Ainsworth, 2006). The pathways are proposed explanations for this study.

The polarization applet

Jones vectors and the Poincaré sphere

For a normalized, fully polarized state,

$$\mathbf{J} = \begin{pmatrix} a \\ b\, e^{i\varphi} \end{pmatrix}, \qquad a, b \geq 0, \quad a^2 + b^2 = 1.$$

The applet uses

$$\mathbf{S} = \begin{pmatrix} a^2 - b^2 \\ 2ab\cos\varphi \\ 2ab\sin\varphi \end{pmatrix}, \qquad \|\mathbf{S}\| = 1.$$

Picture placeholder
Poincaré sphere with
$J_H, J_V, J_D, J_A, J_R, J_L$

A point denotes a polarization state, not a position in the laboratory.

Equations and labels follow the applet's convention.

What the geometric connection does and does not show

$$|\psi\rangle = \cos\frac{\vartheta}{2}|0\rangle + e^{i\varphi}\sin\frac{\vartheta}{2}|1\rangle$$

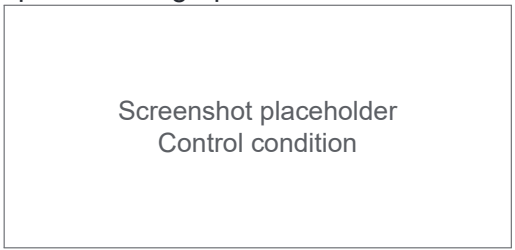
- After choosing a basis and axis convention, pure qubit states and normalized Jones states share the same sphere geometry.
- An overall phase does not change the represented state.
- A lossless wave plate changes the state without reducing intensity.
- A polarizer changes the transmitted state and usually reduces intensity. The sphere alone does not encode that intensity.

The teaching challenge is interpreting a mathematical state space without turning it into a literal physical object.

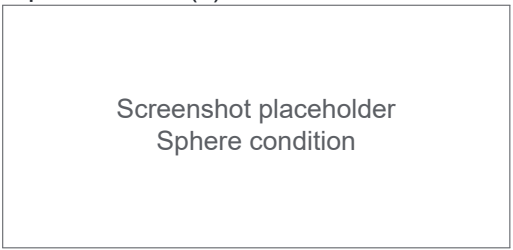
Jones-calculus implementation and the teaching rationale.

Two versions of the applet

Control: Polarization 2
Jones notation and
spatial wave graph.



Sphere: Polarization 1
Adds Poincaré sphere
representation(s).



Both versions use guided parameter changes, hints, solutions, and explanatory feedback. The sphere first appears in step 3 of Polarization 1.

Condition mapping and timing: `app.py` and `challenges.py`.

Hypotheses and study design

Why might the sphere change model understanding?

Functional reasoning, including the architectural profile

For learners with high FG and low FF, the sphere may make functions accessible: wave plates act as **rotations** of the state, while polarizers perform **projections** of the Jones state.

- This support may increase FF and expert alignment, even before literal beliefs change.
- For the dual profile (high FG, high FF), contrasting representations may challenge literal interpretations and reduce FG while FF stays high.

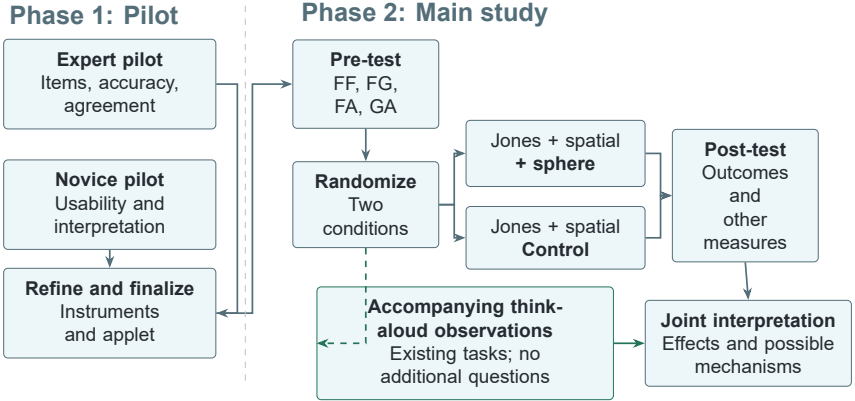
The sphere may support both functional reasoning and reflection on visual fidelity.

Hypotheses: separate belief change from alignment

	Outcome	Sphere versus control, adjusted for baseline
H1a	Functional Fidelity	Higher FF at post-test.
H1b	Fidelity of Gestalt	Lower FG at post-test.
H2	Expert alignment	Higher FA and GA (structural/Gestalt alignment).

Exploratory extension: What spontaneous reasoning during existing applet tasks helps explain learning and difficulty?

Study structure: pilot, then main study



Accompanying think-aloud observations capture reasoning during existing applet tasks.

Phase 1: calibrate the instruments and the applet

Expert pilot

- Classify items as FF, FG, both, or neither.
- Judge factual correctness and the usefulness of analogies.
- Evaluate wording and agreement.

Novice pilot

- Identify confusing controls and instructions.
- Examine interpretation of the representations.
- Assess cognitive load and acceptance.

Refine questionnaire items, applet instructions, and interaction design before the main comparison.

Phase 2: randomized pre/post comparison

Pre-test	Consent, background, prior knowledge; FF, FG, FA, and GA.
Intervention	Random assignment to guided polarization tasks with or without the added sphere.
Post-test	FF, FG, FA, GA; cognitive load, technology acceptance, and spatial reasoning.

Population: second- and third-year physics / related STEM undergraduates.

Target: $N = 218$ recruited; minimum analyzable sample $N = 196$.

Think-aloud observations: investigating mechanisms

Students verbalize their thoughts while working through the existing applet tasks, without additional questions or activities.

- Observe how students attempt to incorporate the Poincaré sphere into their mental model.
- Trace difficulties in connecting spatial, symbolic, and sphere representations.
- Examine spontaneously expressed conflicts between literal interpretations and functional relationships.
- Follow how these conflicts are resolved, maintained, or bypassed during task engagement.

Time-ordered observations can provide evidence for mechanisms underlying learning and difficulty.

Bring quantitative outcomes and verbal reasoning together

Quantitative strand

- Estimate condition effects on post-test FF, FG, FA, and GA.
- Adjust for baseline scores and spatial reasoning.
- Explore initial FG, spatial reasoning, and cognitive load.

Qualitative strand

- Trace interpretation, comparison, and revision episodes.
- Examine spontaneously expressed realism-related tension.
- Examine alternative explanations and contrasting cases.

The experiment estimates condition effects. Observations help explain individual learning episodes.

Analysis and current status

Planned primary analysis: MANCOVA

Joint outcomes

$$\mathbf{Y}_{\text{post}} = \begin{pmatrix} FF_{\text{post}} \\ FG_{\text{post}} \\ FA_{\text{post}} \\ GA_{\text{post}} \end{pmatrix}$$

- **Factor:** condition, with or without the additional Poincaré sphere.
- **Covariates:** baseline outcome scores and spatial reasoning ability.
- **Multivariate test:** assess the adjusted condition effect across the four outcomes.
- **Outcome-specific analyses:** examine the directional predictions of H1a, H1b, and H2.

Exploratory analyses: H3 and H4

H3: Spatial reasoning and cognitive load

In the planned experimental tracking subset, regress pre/post changes in the study outcomes on spatial reasoning ability and intrinsic, extraneous, and germane cognitive load.

Do these characteristics help explain variation in learning?

H4: Initial Fidelity of Gestalt as a moderator

Examine the interaction between baseline FG and condition in predicting post-test FF or FA, controlling for the corresponding baseline outcome.

Does the effect of the additional visualization depend on students' initial beliefs about visual fidelity?

Initial pilot findings

Pilot studies are ongoing.

Expert pilot

- Add sample and completion status.
- Add item-classification and agreement findings.
- Add resulting item revisions.

Novice pilot

- Add sample and completion status.
- Add interpretation or usability issues.
- Add resulting applet revisions.

Placeholder: pilot figure, applet screenshot, or anonymized spontaneous verbalization

What would the findings tell us?

Possible pattern	Interpretation to examine
Higher FA / GA; little FF / FG change	Improved expert alignment within model understanding, with stable fidelity beliefs.
Higher FF; lower FG	Consistent with a more functional stance toward models.
Spontaneous tension followed by revision	Time-ordered evidence consistent with a proposed learning mechanism.
Confusion or higher load without gains	Difficulties in connecting representations may limit learning.

Interpretive scenarios, not empirical results.

Contribution of the polarization study

- Examine fidelity beliefs and expert alignment as aspects of model understanding.
- Test whether the additional sphere supports functional reasoning and changes visual-fidelity beliefs.
- Use spontaneous think-aloud observations to investigate learning and difficulty during existing tasks.

Scope of the conclusions

No additional interview tasks or follow-up questions. No direct measurement of model competencies.

Can an additional representation change both what learners understand and what they think a model is for?

Appendix: the six reference states

State	Normalized Jones vector	Applet coordinates (S_1, S_2, S_3)
Horizontal J_H	$(1,0)^{\mathsf{T}}$	$(1,0,0)$
Vertical J_V	$(0,1)^{\mathsf{T}}$	$(-1,0,0)$
Diagonal J_D	$(1,1)^{\mathsf{T}}/\sqrt{2}$	$(0,1,0)$
Antidiagonal J_A	$(1,-1)^{\mathsf{T}}/\sqrt{2}$	$(0,-1,0)$
Right-circular J_R	$(1,i)^{\mathsf{T}}/\sqrt{2}$	$(0,0,1)$
Left-circular J_L	$(1,-i)^{\mathsf{T}}/\sqrt{2}$	$(0,0,-1)$

Labels and phase convention follow the applet. Overall phase is physically irrelevant.

The learning sequence

Steps	Topic	Learner activity
1–3	Polarization states	Change amplitudes and relative phase to construct linear and circular states.
4–6	Wave plates	Vary retardance and fast-axis orientation to transform a state.
7	Linear polarizers	Relate transmission-axis orientation to output intensity.
8–9	Combined operators	Coordinate wave plates and polarizers to meet a target.

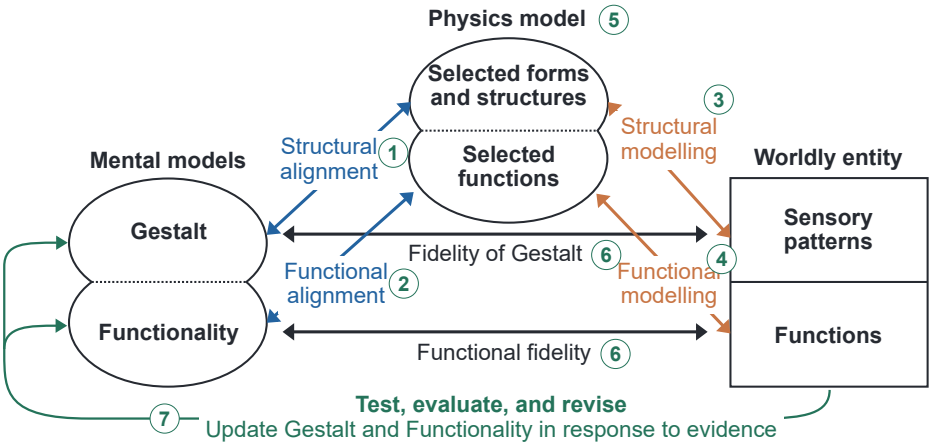
Nine challenges precede a completion screen and survey return link.

Challenge definitions. A revealed solution can also satisfy the target.

Appendix: measures and protocol decisions

- Cognitive load: (Krieglstein et al., 2023); technology acceptance: (Abdullah and Ward, 2016); spatial reasoning: (Young et al., 2025).
- Specify observation sampling, recording, consent, and coding before collection. Observe existing tasks without added prompts, activities, or follow-up questions.
- Reconcile the stated covariate count, missing spatial-reasoning data, and the final primary-analysis specification.
- Retain the preregistered completion exclusions. Clarify how the primary model includes SRA while retaining participants with missing SRA.
- Match tutorial wording and starting states; prevent control-group sphere exposure before the post-test; verify participant matching and timing records.

Appendix: competencies as theoretical context



Green markers: competencies, explained on the next slide. The return loop makes evidence-based revision explicit.

Appendix: examples of modeling practices

	Competency	Illustrative practice, not an added task
1	Structural mapping	Relate Jones components, the wave graph, and sphere coordinates.
2	Functional mapping	Explain how a phase change appears in each representation.
3	Structural construction	Choose a drawing that makes a relevant feature visible.
4	Functional modelling and testing	Predict a wave plate's effect, then test it.
5	Selection and transformation	Choose a representation and translate the same state.
6	Metacognitive reflection	Explain which visual features are conventional or nonliteral.
7	Evidence-based revision	Revise the prediction or representation after a mismatch.

Conceptual examples from the supplied extension. The study does not assess model competencies.

References I

Abdullah, F. and R. Ward (2016). “Developing a General Extended Technology Acceptance Model for E-Learning (GETAMEL) by analysing commonly used external factors”. In: *Computers in Human Behavior* 56, pp. 238–256.

Ainsworth, Shaaron (2006). “DeFT: A conceptual framework for considering learning with multiple representations”. In: *Learning and Instruction* 16.3, pp. 183–198. DOI: 10.1016/j.learninstruc.2006.03.001.

Krieglstein, Felix, Maik Beege, Günter Daniel Rey, Christina Sanchez-Stockhammer, and Sascha Schneider (2023). “Development and Validation of a Theory-Based Questionnaire to Measure Different Types of Cognitive Load”. In: *Educational Psychology Review* 35.1, p. 9. DOI: 10.1007/s10648-023-09738-0.

References II

Schwarz, Christina V., Brian J. Reiser, Elizabeth A. Davis, Lisa Kenyon, Andres Achér, David Fortus, Yael Shwartz, Barbara Hug, and Joe Krajcik (2009). “Developing a learning progression for scientific modeling: Making scientific modeling accessible and meaningful for learners”. In: *Journal of Research in Science Teaching* 46.6, pp. 632–654. DOI: 10.1002/tea.20311.

Ubben, Malte S. and Philipp Bitzenbauer (2022). “Two Cognitive Dimensions of Students’ Mental Models in Science: Fidelity of Gestalt and Functional Fidelity”. In: *Education Sciences* 12.3, p. 163. DOI: 10.3390/educsci12030163.

Young, S. R., J. Kim, K. McKee, D. R. Doyle, M. A. Novack, W. Revelle, R. Gershon, and E. M. Dworak (2025). “Validation of International Cognitive Ability Resource (ICAR) Implemented in Mobile Toolbox (MTB)”. In: *Journal of Intelligence* 13.12, p. 154. DOI: 10.3390/jintelligence13120154.